Joshua Faskowitz

Postdoctoral Fellow | Network Neuroscience & Functional MRI

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Research Profile

I study how large-scale brain organization forms from interactions among brain areas, with an emphasis on edge-centric models, mesoscale structure, and the interpretation of functional MRI signals. My work blends cognitivie neuroscience with network science, computational modeling, and neuroimaging analysis to further our understanding about how brain dynamics can explain behavior and congition.

Education

Indiana University Bloomington — *Bloomington, IN*
Ph.D. in Neuroscience and Psychology — *2021*
Advisor: Dr. Olaf Sporns
Committee: Dr. Olaf Sporns, Dr. Aina Puce, Dr. Amanda Mejia, Dr. Richard Betzel

University of Southern California — *Los Angeles, CA*
B.A. in Neuroscience, Cognitive Science — *2014*

Research Experience

Postdoctoral Fellow — *Bethesda, MD*
Section on Functional Imaging Methods, National Institute of Mental Health — *2022 – present*
PI: Dr. Peter A. Bandettini

Postdoctoral Researcher — *Bloomington, IN*
Indiana University Department of Psychological and Brain Sciences — *2021 – 2022*
PI: Dr. Olaf Sporns and Dr. Richard Betzel

Graduate Student — *Bloomington, IN*
IU Computational Cognitive Neuroscience Lab — *2016 – 2021*
PI: Dr. Olaf Sporns

Project Assistant — *Los Angeles, CA*
USC Imaging Genetics Center — *2014 – 2016*
PI: Dr. Paul Thompson; under supervision of Dr. Neda Jahanshad

Undergraduate Research Assistant — *Los Angeles, CA*
USC Emotion & Cognition Lab — *2012 – 2014*
PI: Dr. Mara Mather; under supervision of Dr. Allison Ponzio

Research Keywords

Network neuroscience; network science algorithms and applications; community detection; edge-centric modeling; fMRI analysis and interpretation; time-varying connectivity

Honors & Awards

IU Department of Psychological and Brain Sciences Irving J. Saltzman Award — *2023*
Departmental outstanding recent PhD graduate ($2,000)

IU Gill Center Linda and Jack Gill Graduate Student Award Honorable Mention — *2021*
One of four awardees selected from regional neuroscience programs at IUB, IUPUI, Purdue, and Notre Dame

IU Department of Psychological and Brain Sciences J. R. Kantor Graduate Award *2021*
Departmental outstanding advanced graduate student; shared with another student ($1,750)

IU College of Arts and Sciences Dissertation Research Fellowship *2020-2021*
$20,000

NSF Graduate Research Fellowship Program *2016-2019*
$34,000 per year for 3 years

IU Department of Psychological and Brain Sciences Rebec Fellow *2016*
$4,000

USC Trustee Scholarship *2010-2014*
Full tuition awarded for academics, leadership, and community service ($43,000 per year)

USC Order of Troy *2014*
For academic and leadership excellence; one of 112 undergraduate awardees

Renaissance Scholar and Discovery Scholar *2014*
For excelling in two widely separated fields of study and for conducting scholarly research; one of 64 graduates with multiple scholar honors at graduation

Invited Talks

Brain connectivity and advanced analysis strategies *2024, 2026*
fMRI Summer Course talk series at NIH

Spatial Similarity Maps Across the Cortex *2026*
OHBM 2026 oral session

Network Outcomes, Uses, and Null Models *2022*
Practical Applications of Network Neuroscience Workshop, co-organized with Dr. Jenya Chumin

Brain Networks: Close to the Edge *2022*
Washington University in St. Louis Neuroimaging in Health and Disease seminar series

Brain Networks: Close to the Edge *2022*
Dr. Lucina Uddin lab meeting

fMRI Preprocessing Workshop *2022*
Three days covering workflows for processing neuroimaging data using high-performance computing

Publicly Available Neuroimaging Databases *2020*
CUNY 2020 workshop on remote data collection

Teaching

The Consciousness Network *2023, 2024, 2026*
NIH summer journal club, co-organized with Dr. Sharif Kronemer

P303: Health Psychology *2019*
Teaching assistant

P211: Methods of Experimental Psychology *2018*
Instructor of record

Workshops & Hackathons Attended

NIMH Grant Writing Workshop *2024, 2025*
Full-day overview of grantsmanship

Brainhack-Networks *2019*
Pre-NetSci 2019, University of Vermont

Brainhack Global at IU *2018*
Indiana University

Indiana Neuroimaging Symposium and Hackathon *2016, 2017*
Purdue University

Neurohackweek *2017*
University of Washington eScience Institute

OHBM Hackathon *2015*
Pre-OHBM 2015, Honolulu, U.S.A.

Code & Data Sharing Examples

Multi Atlas Transfer Tools
Fit surface parcellations; run more than 20,000 times as an app on Brainlife.io

fMRI-2-mat
Denoising and matrix-based downstream analysis of fMRI data

Lifespan structural connectivity matrices from Faskowitz (2018)
Shared dataset downloaded >200 times.

Brain-networks-across-the-web
Compilation of openly available brain networks

Research Toolbox

Coding and Scripting: MATLAB (expert), shell scripting (bash, awk, grep, sed) (expert), Python (competent), R (competent), LaTeX (basic)

Computational Tools: GitHub, HPC environments, Docker/Singularity, Unix

Neuroimaging Packages: FSL, FreeSurfer, ANTs, Dipy, fMRIPrep, MRtrix, AFNI

Journal Peer Review

Please see Web of Science profile (K-1330-2019) for list of reviewer activity, including:

Aperture Neuro; Brain Imaging and Behavior; Brain Structure and Function; Biological Psychiatry; Cerebral Cortex; Human Brain Mapping; Nature Communications; Nature Neuroscience; Network Neuroscience; NeuroImage; Scientific Data; Scientific Reports; The Journal of Neuroscience; Trends in Cognitive Sciences

Professional Affiliations

Organization for Human Brain Mapping; Society for Neuroscience; Network Science Society

Preprints

10. **The Neural Code of Neuroticism**
    Johanna L Popp, Martin Weiß, **Joshua Faskowitz**, Kirsten Hilger
    bioRxiv (2026)
9. **Resting-State fMRI and the Risk of Overinterpretation: Noise, Mechanisms, and a Missing Rosetta Stone**
    Gang Chen, Zhengchen Cai, Konrad P Kording, Thomas T Liu, **Joshua Faskowitz**, Peter A Bandettini, Bharat Biswal, Paul A Taylor
    bioRxiv (2025)
8. **Connectome architecture favours within-module diffusion and between-module routing**
    Caio Seguin, Maria Grazia Puxeddu, **Joshua Faskowitz**, Richard F Betzel, Olaf Sporns
    bioRxiv (2025)
7. **Commentary on Pang et al.(2023) nature**
    Joshua Faskowitz, Daniel Moyer, Daniel Handwerker, Javier Gonzalez-Castillo, Peter Bandettini, Saad

Jbabdi, Richard Betzel
bioRxiv (2023)

6. **Stability of brain-behavior correlation patterns across measures of social support**
Haily Merritt, **Joshua Faskowitz**, Marlen Z Gonzalez, Richard F Betzel
bioRxiv (2023)
5. **System-level high-amplitude co-fluctuations**
Richard Betzel, Evgeny J Chumin, Farnaz Zamani Esfahlani, Jacob Tanner, **Joshua Faskowitz**
bioRxiv (2022)
4. **Multi-policy models of interregional communication in the human connectome**
Richard Betzel, **Joshua Faskowitz**, Bratislav Misic, Olaf Sporns, Caio Seguin
bioRxiv (2022)
3. **An edge-centric model for harmonizing multi-relational network datasets**
Joshua Faskowitz, Jacob Tanner, Bratislav Misic, Richard Betzel
bioRxiv (2021)
2. **Do Candidate Genes Affect the Brain's White Matter Microstructure? Large-Scale Evaluation of 6,165 Diffusion MRI Scans**
Neda Jahanshad, Habib Ganjgahi, Janita Bralten, Anouk Den Braber, **Joshua Faskowitz**, Annchen Knodt, Herve Lemaitre, Talia Nir, Binish Patel, Stuart Richie, et al.
BioRxiv (2017)
1. **An Empirical Study of Continuous Connectivity Degree Sequence Equivalents**
Daniel Moyer, Boris A. Gutman, **Joshua Faskowitz**, Neda Jahanshad, Paul M. Thompson
arXiv (2016)

Journal Articles

65. **Human cortex organizes dynamic co-fluctuations along the sensorimotor-association axis**
De-Zhi Jin, Changsong Zhou, Xi-Nian Zuo, **Joshua Faskowitz**, Ting Xu, Ye He
Nature Communications (2026)
64. **Trait-Relevant Tasks Improve Personality Prediction From Structural-Functional Brain Network Coupling**
Johanna L Popp, Jonas A Thiele, **Joshua Faskowitz**, Caio Seguin, Olaf Sporns, Kirsten Hilger
Human Brain Mapping 47(7) e70533 (2026)
63. **Quantifying the compressibility of the human brain**
Nicholas J Weaver, **Joshua Faskowitz**, Richard F Betzel, Christopher W Lynn
Proceedings of the National Academy of Sciences 123(4) e2531115123 (2026)
62. **Controlling the human connectome with spatially diffuse input signals**
Richard Betzel, Maria Grazia Puxeddu, Caio Seguin, Vincent Bazinet, Andrea Luppi, Alina Podschun, S Parker Singleton, **Joshua Faskowitz**, Vibin Parakkattu, Bratislav Misic, et al.
Communications biology (2026)
61. **Dynamic adaptation to novelty in the brain is related to arousal and intelligence**
Jacob Tanner, **Joshua Faskowitz**, Daniel P Kennedy, Richard F Betzel
Cerebral Cortex 36(4) bhag025 (2026)
60. **Mapping high-amplitude fMRI edge time series events across space and time**
Joshua Faskowitz, Peter A Bandettini, Javier Gonzalez-Castillo
Imaging Neuroscience (2026)
59. **Leveraging multivariate information for community detection in functional brain networks**
Maria Grazia Puxeddu, Maria Pope, Thomas F Varley, **Joshua Faskowitz**, Olaf Sporns
Communications Biology 8(1) 840 (2025)
58. **Structural-functional brain network coupling during cognitive demand reveals intelligence-relevant communication strategies**
Johanna L Popp, Jonas A Thiele, **Joshua Faskowitz**, Caio Seguin, Olaf Sporns, Kirsten Hilger
Communications Biology 8(1) 855 (2025)

57. **Variation in high-amplitude events across the human lifespan**
Youngheun Jo, Jacob Tanner, Caio Seguin, **Joshua Faskowitz**, Richard F Betzel
Network Neuroscience 1-27 (2025)

56. **Decoding the human brain during intelligence testing**
Jonas A Thiele, **Joshua Faskowitz**, Olaf Sporns, Adam Chuderski, Rex Jung, Kirsten Hilger
Communications Biology (2025)

55. **Time-varying synergy/redundancy dominance in the human cerebral cortex**
Maria Pope, Thomas F Varley, Maria Grazia Puxeddu, **Joshua Faskowitz**, Olaf Sporns
Journal of Physics: Complexity 6(1) 015015 (2025)

54. **Choosing explanation over performance: Insights from machine learning-based prediction of human intelligence from brain connectivity**
Jonas A Thiele, **Joshua Faskowitz**, Olaf Sporns, Kirsten Hilger
PNAS nexus 3(12) pgae519 (2024)

53. **Stability and variation of brain-behavior correlation patterns across measures of social support**
Haily Merritt, **Joshua Faskowitz**, Marlen Z Gonzalez, Richard F Betzel
Imaging Neuroscience 2 1-18 (2024)

52. **A multi-modal, asymmetric, weighted, and signed description of anatomical connectivity**
Jacob Tanner, **Joshua Faskowitz**, Andreia Sofia Teixeira, Caio Seguin, Ludovico Coletta, Alessandro Gozzi, Bratislav Mišić, Richard F Betzel
Nature communications 15(1) 5865 (2024)

51. **Reliability of variability and complexity measures for task and task-free BOLD fMRI.**
MH Wehrheim, **J Faskowitz**, AL Schubert, CJ Fiebach
Human Brain Mapping 45(10) e26778-e26778 (2024)

50. **Neuroanatomical changes observed over the course of a human pregnancy**
Laura Pritschet, Caitlin M Taylor, Daniela Cossio, **Joshua Faskowitz**, Tyler Santander, Daniel A Handwerker, Hannah Grotzinger, Evan Layher, Elizabeth R Chrastil, Emily G Jacobs
Nature Neuroscience 27(11) 2253-2260 (2024)

49. **Functional hierarchies in brain dynamics characterized by signal reversibility in ferret cortex**
Sebastian Idesis, Sebastian Geli, **Joshua Faskowitz**, Jakub Vohryzek, Yonatan Sanz Perl, Florian Pieper, Edgar Galindo-Leon, Andreas K Engel, Gustavo Deco
PLOS Computational Biology 20(1) e1011818 (2024)

48. **Modular subgraphs in large-scale connectomes underpin spontaneous co-fluctuation events in mouse and human brains**
Elisabeth Ragone, Jacob Tanner, Youngheun Jo, Farnaz Zamani Esfahlani, **Joshua Faskowitz**, Maria Pope, Ludovico Coletta, Alessandro Gozzi, Richard Betzel
Communications Biology 7(1) 126 (2024)

47. **Relation of connectome topology to brain volume across 103 mammalian species**
Maria Grazia Puxeddu, **Joshua Faskowitz**, Caio Seguin, Yossi Yovel, Yaniv Assaf, Richard Betzel, Olaf Sporns
Plos Biology 22(2) e3002489 (2024)

46. **brainlife. io: a decentralized and open-source cloud platform to support neuroscience research**
Soichi Hayashi, Bradley A Caron, Anibal Sólon Heinsfeld, Sophia Vinci-Booher, Brent McPherson, Daniel N Bullock, Giulia Bertò, Guiomar Niso, Sandra Hanekamp, ..., **Joshua Faskowitz**, ..., Franco Pestilli
Nature methods 1-5 (2024)

45. **Structural-functional brain network coupling predicts human cognitive ability**
Johanna L Popp, Jonas A Thiele, **Joshua Faskowitz**, Caio Seguin, Olaf Sporns, Kirsten Hilger
NeuroImage 290 120563 (2024)

44. **Multivariate information theory uncovers synergistic subsystems of the human cerebral cortex**
Thomas F Varley, Maria Pope, **Joshua Faskowitz**, Olaf Sporns
Communications biology 6(1) 451 (2023)

43. **Partial entropy decomposition reveals higher-order information structures in human brain activity**

Thomas F Varley, Maria Pope, Puxeddu Maria Grazia, **Joshua Faskowitz**, Olaf Sporns
Proceedings of the National Academy of Sciences 120(30) e2300888120 (2023)

42. **Few temporally distributed brain connectivity states predict human cognitive abilities**
Maren H Wehrheim, **Joshua Faskowitz**, Olaf Sporns, Christian J Fiebach, Matthias Kaschube, Kirsten Hilger
NeuroImage 277 120246 (2023)

41. **Co-evolving dynamics and topology in a coupled oscillator model of resting brain function**
Maria Pope, Caio Seguin, Thomas F Varley, **Joshua Faskowitz**, Olaf Sporns
NeuroImage 277 120266 (2023)

40. **Living on the edge: network neuroscience beyond nodes**
Richard F Betzel, **Joshua Faskowitz**, Olaf Sporns
Trends in cognitive sciences 27(11) 1068 (2023)

39. **A low dimensional embedding of brain dynamics enhances diagnostic accuracy and behavioral prediction in stroke**
Sebastian Idesis, Michele Allegra, Jakub Vohryzek, Yonatan Sanz Perl, **Joshua Faskowitz**, Olaf Sporns, Maurizio Corbetta, Gustavo Deco
Scientific Reports 13(1) 15698 (2023)

38. **High-amplitude network co-fluctuations linked to variation in hormone concentrations over the menstrual cycle**
Sarah Greenwell, **Joshua Faskowitz**, Laura Pritschet, Tyler Santander, Emily G Jacobs, Richard F Betzel
Network Neuroscience 7(3) 1181-1205 (2023)

37. **Hierarchical organization of spontaneous co-fluctuations in densely sampled individuals using fMRI**
Richard F Betzel, Sarah A Cutts, Jacob Tanner, Sarah A Greenwell, Thomas Varley, **Joshua Faskowitz**, Olaf Sporns
Network Neuroscience 7(3) 926-949 (2023)

36. **Synchronous high-amplitude co-fluctuations of functional brain networks during movie-watching**
Jacob C Tanner, **Joshua Faskowitz**, Lisa Byrge, Daniel P Kennedy, Olaf Sporns, Richard F Betzel
Imaging Neuroscience 1 1-21 (2023)

35. **Connectome topology of mammalian brains and its relationship to taxonomy and phylogeny**
Joshua Faskowitz, Maria Grazia Puxeddu, Martijn P van den Heuvel, Bratislav Mišić, Yossi Yovel, Yaniv Assaf, Richard F Betzel, Olaf Sporns
Frontiers in neuroscience 16 (2022)

34. **Multi-modal and multi-subject modular organization of human brain networks**
Maria Grazia Puxeddu, **Joshua Faskowitz**, Olaf Sporns, Laura Astolfi, Richard F Betzel
NeuroImage 264 119673 (2022)

33. **Multitask Brain Network Reconfiguration Is Inversely Associated with Human Intelligence.**
JA Thiele, **J Faskowitz**, O Sporns, K Hilger
Cerebral Cortex (New York, NY: 1991) (2022)

32. **Effects of diffusion signal modeling and segmentation approaches on subthalamic nucleus parcellation**
Demetrio Milardi, Gianpaolo Antonio Basile, **Joshua Faskowitz**, Salvatore Bertino, Angelo Quartarone, Giuseppe Pio Anastasi, Alessia Bramanti, Rosella Ciurleo, Alberto Cacciola
NeuroImage 250 118959 (2022)

31. **Cortico-subcortical interactions in overlapping communities of edge functional connectivity**
Evgeny J Chumin, **Joshua Faskowitz**, Farnaz Zamani Esfahlani, Youngheun Jo, Haily Merritt, Jacob Tanner, Sarah A Cutts, Maria Pope, Richard Betzel, Olaf Sporns
NeuroImage 250 118971 (2022)

30. **Local structure-function relationships in human brain networks across the lifespan**
Farnaz Zamani Esfahlani, **Joshua Faskowitz**, Jonah Slack, Bratislav Mišić, Richard F Betzel
Nature communications 13(1) 1-16 (2022)

29. **Individualized event structure drives individual differences in whole-brain functional connectivity**
Richard F Betzel, Sarah A Cutts, Sarah Greenwell, **Joshua Faskowitz**, Olaf Sporns
NeuroImage 252 118993 (2022)

28. **Edge-centric analysis of stroke patients: An alternative approach for biomarkers of lesion recovery**
Sebastian Idesis, **Joshua Faskowitz**, Richard F Betzel, Maurizio Corbetta, Olaf Sporns, Gustavo Deco
NeuroImage: Clinical 103055 (2022)

27. **Uncovering individual differences in fine-scale dynamics of functional connectivity**
Sarah A Cutts, **Joshua Faskowitz**, Richard F Betzel, Olaf Sporns
Cerebral Cortex (New York, NY: 1991) (2022)

26. **Edges in brain networks: contributions to models of structure and function** link
Joshua Faskowitz, Richard F Betzel, Olaf Sporns
Network Neuroscience 1-63 (2021-08)

25. **The diversity and multiplexity of edge communities within and between brain systems**
Youngheun Jo, Farnaz Zamani Esfahlani, **Joshua Faskowitz**, Evgeny J Chumin, Olaf Sporns, Richard F Betzel
Cell reports 37(7) 110032 (2021)

24. **Modularity maximization as a flexible and generic framework for brain network exploratory analysis**
Farnaz Zamani Esfahlani, Youngheun Jo, Maria Grazia Puxeddu, Haily Merritt, Jacob C Tanner, Sarah Greenwell, Riya Patel, **Joshua Faskowitz**, Richard F Betzel
NeuroImage 244 118607 (2021)

23. **Mapping individual differences across brain network structure to function and behavior with connectome embedding**
Gidon Levakov, **Joshua Faskowitz**, Galia Avidan, Olaf Sporns
NeuroImage 118469 (2021)

22. **Subject identification using edge-centric functional connectivity**
Youngheun Jo, **Joshua Faskowitz**, Farnaz Zamani Esfahlani, Olaf Sporns, Richard F Betzel
NeuroImage 118204 (2021)

21. **Collegiate athlete brain data for white matter mapping and network neuroscience**
Bradley Caron, Ricardo Stuck, Brent McPherson, Daniel Bullock, Lindsey Kitchell, **Joshua Faskowitz**, Derek Kellar, Hu Cheng, Sharlene Newman, Nicholas Port, et al.
Scientific Data 8(1) 1-17 (2021)

20. **Dynamic expression of brain functional systems disclosed by fine-scale analysis of edge time series**
Olaf Sporns, **Joshua Faskowitz**, Andreia Sofia Teixeira, Sarah A Cutts, Richard F Betzel
Network Neuroscience 5(2) 405-433 (2021)

19. **Association of immunosuppression and viral load with subcortical brain volume in an international sample of people living with HIV**
Talia M Nir, Jean-Paul Fouche, Jintanat Ananworanich, Beau M Ances, Jasmina Boban, Bruce J Brew, Linda Chang, Joga R Chaganti, Christopher RK Ching, …, **J Faskowitz**, …, N Jahanshad
JAMA network open 4(1) e2031190-e2031190 (2021)

18. **The modular organization of brain cortical connectivity across the human lifespan**
Maria Grazia Puxeddu, **Joshua Faskowitz**, Richard F Betzel, Manuela Petti, Laura Astolfi, Olaf Sporns
NeuroImage 116974 (2020)

17. **Aging relates to a disproportionately weaker functional architecture of brain networks during rest and task states**
Colleen Hughes, **Joshua Faskowitz**, Brittany S Cassidy, Olaf Sporns, Anne C Krendl
NeuroImage 116521 (2020)

16. **High-amplitude cofluctuations in cortical activity drive functional connectivity**
Farnaz Zamani Esfahlani, Youngheun Jo, **Joshua Faskowitz**, Lisa Byrge, Daniel P Kennedy, Olaf Sporns, Richard F Betzel
Proceedings of the National Academy of Sciences 117(45) 28393-28401 (2020)

15. **Edge-centric functional network representations of human cerebral cortex reveal overlapping system-level architecture**

Joshua Faskowitz, Farnaz Zamani Esfahlani, Youngheun Jo, Olaf Sporns, Richard F Betzel
Nature Neuroscience (2020)

14. **The reliability and heritability of cortical folds and their genetic correlations across hemispheres**
Fabrizio Pizzagalli, Guillaume Auzias, Qifan Yang, Samuel R Mathias, **Joshua Faskowitz**, Joshua D Boyd, Armand Amini, Denis Rivière, Katie L McMahon, Greig I de Zubicaray, et al.
Communications Biology 3(1) 1-12 (2020)

13. **Mapping the community structure of the rat cerebral cortex with weighted stochastic block modeling**
Joshua Faskowitz, Olaf Sporns
Brain Structure and Function 225(1) 71-84 (2020)

12. **MULTI-SITE META-ANALYSIS OF MORPHOMETRY**
Neda Jahanshad, **Joshua I Faskowitz**, Gennady Roshchupkin, Derrek Hibar, Boris A Gutman, Nicholas J Tustison, Hieab HH Adams, Wiro Niessen, Meike W Vernooij, M Arfan Ikram, et al.
IEEE/ACM transactions on computational biology and bioinformatics (2019)

11. **Age differences in specific neural connections within the Default Mode Network underlie theory of mind**
Colleen Hughes, Brittany S Cassidy, **Joshua Faskowitz**, Andrea Avena-Koenigsberger, Olaf Sporns, Anne C Krendl
NeuroImage 191 269-277 (2019)

10. **Weighted Stochastic Block Models of the Human Connectome across the Life Span**
Joshua Faskowitz, Xiaoran Yan, Xi-Nian Zuo, Olaf Sporns
Scientific reports 8(1) 12997 (2018)

9. **Cortical brain abnormalities in 4474 individuals with schizophrenia and 5098 control subjects via the Enhancing Neuro Imaging Genetics Through Meta Analysis (ENIGMA) Consortium**
Theo GM Van Erp, Esther Walton, Derrek P Hibar, Lianne Schmaal, Wenhao Jiang, David C Glahn, Godfrey D Pearlson, Nailin Yao, ..., **J Faskowitz**, ..., J. A. Turner
Biological psychiatry 84(9) 644-654 (2018)

8. **Systemic inflammation as a predictor of brain aging: contributions of physical activity, metabolic risk, and genetic risk**
Fabian Corlier, George Hafzalla, **Joshua Faskowitz**, Lewis H Kuller, James T Becker, Oscar L Lopez, Paul M Thompson, Meredith N Braskie
Neuroimage 172 118-129 (2018)

7. **Widespread white matter microstructural differences in schizophrenia across 4322 individuals: results from the ENIGMA Schizophrenia DTI Working Group**
Sinead Kelly, Neda Jahanshad, A Zalesky, P Kochunov, Ingrid Agartz, C Alloza, OA Andreassen, C Arango, N Banaj, ..., **J Faskowitz**, ..., G Donohoe
Molecular psychiatry 23(5) 1261 (2018)

6. **Continuous representations of brain connectivity using spatial point processes**
Daniel Moyer, Boris A Gutman, **Joshua Faskowitz**, Neda Jahanshad, Paul M Thompson
Medical image analysis 41 32-39 (2017)

5. **Diverging volumetric trajectories following pediatric traumatic brain injury**
Emily L Dennis, **Joshua Faskowitz**, Faisal Rashid, Talin Babikian, Richard Mink, Christopher Babbitt, Jeffrey Johnson, Christopher C Giza, Neda Jahanshad, Paul M Thompson, et al.
Neuroimage: clinical 15 125-135 (2017)

4. **Relationship of a common OXTR gene variant to brain structure and default mode network function in healthy humans**
Junping Wang, Meredith N Braskie, George W Hafzalla, **Joshua Faskowitz**, Katie L McMahon, Greig I de Zubicaray, Margaret J Wright, Chunshui Yu, Paul M Thompson
Neuroimage 147 500-506 (2017)

3. **Partitioning heritability analysis reveals a shared genetic basis of brain anatomy and schizophrenia**
Phil H Lee, Justin T Baker, Avram J Holmes, Neda Jahanshad, Tian Ge, Jae-Yoon Jung, Yanela Cruz, Dara

S Manoach, Derrek P Hibar, **Joshua Faskowitz**, et al.
Molecular psychiatry 21(12) 1680 (2016)

2. **Subcortical volumetric abnormalities in bipolar disorder**
DP Hibar, Lars T Westlye, Theo GM van Erp, J Rasmussen, Cassandra D Leonardo, **J Faskowitz**, Unn K Haukvik, Cecilie Bhandari Hartberg, Nhat Trung Doan, Ingrid Agartz, et al.
Molecular psychiatry 21(12) 1710 (2016)

1. **Heritability and reliability of automatically segmented human hippocampal formation subregions**
Christopher D Whelan, Derrek P Hibar, Laura S van Velzen, Anthony S Zannas, Tania Carrillo-Roa, Katie McMahon, Gautam Prasad, Sinéad Kelly, **Joshua Faskowitz**, Greig deZubiracay, et al.
Neuroimage 128 125-137 (2016)

Peer-Reviewed Conference Proceedings

25. **Genetic Connectivity–Correlated Genetic Control of Cortical Thickness, Brain Volume, and White Matter**
Daniel A Rinker, Neda Jahanshad, Derrek P Hibar, **Joshua Faskowitz**, Katie L McMahon, Greig I de Zubicaray, Margaret J Wright, Paul M Thompson
Imaging Genetics; Academic Press; pp. 25-43 (2018)

24. **Sulcal-based morphometry in Parkinson's disease: a study of reliability and disease effects**
Fabrizio Pizzagalli, Guillaume Auzias, Armand Amini, **Joshua Faskowitz**, Faisal Rashid, Daniel Moyer, Peter Kochunov, Denis Rivière, Jean-François Mangin, Paul M Thompson, et al.
14th International Symposium on Medical Information Processing and Analysis; International Society for Optics and Photonics; Vol. 10975, pp. 109750T (2018)

23. **Voxelwise meta-analysis of brain structural associations with genome-wide polygenic risk for Alzheimer's disease**
Linda Ding, Alyssa H Zhu, Arvin Saremi, **Joshua I Faskowitz**, Asta Håberg, Paul M Thompson, Neda Jahanshad
14th International Symposium on Medical Information Processing and Analysis; International Society for Optics and Photonics; Vol. 10975, pp. 109750L (2018)

22. **Multisite metaanalysis of image-wide genome-wide associations with morphometry**
Neda Jahanshad, Gennady Roshchupkin, **Joshua Faskowitz**, Derrek P Hibar, Boris A Gutman, Hieab HH Adams, Wiro J Niessen, Meike W Vernooij, M Arfan Ikram, Marcel P Zwiers, et al.
Imaging Genetics; Academic Press; pp. 1-23 (2018)

21. **Simultaneous Matrix Diagonalization for Structural Brain Networks Classification**
Nikita Mokrov, Maxim Panov, Boris A Gutman, **Joshua I Faskowitz**, Neda Jahanshad, Paul M Thompson
International Conference on Complex Networks and their Applications; Springer, Cham; pp. 1261-1270 (2017)

20. **Classifying phenotypes based on the community structure of human brain networks**
Anvar Kurmukov, Marina Ananyeva, Yulia Dodonova, Boris Gutman, **Joshua Faskowitz**, Neda Jahanshad, Paul Thompson, Leonid Zhukov
Graphs in Biomedical Image Analysis, Computational Anatomy and Imaging Genetics; Springer, Cham; pp. 3-11 (2017)

19. **Mapping age effects along fiber tracts in young adults**
Emily L Dennis, Faisal Rashid, **Josh Faskowitz**, Yan Jin, Katie L McMahon, Greig I De Zubicaray, Nicholas G Martin, Ian B Hickie, Margaret J Wright, Neda Jahanshad, et al.
2017 IEEE 14th International Symposium on Biomedical Imaging (ISBI 2017); IEEE; pp. 101-104 (2017)

18. **A comparison of network definitions for detecting sex differences in brain connectivity using Support Vector Machines**
George W Hafzalla, Anjanibhargavi Ragothaman, **Joshua Faskowitz**, Neda Jahanshad, Katie L McMahon, Greig I De Zubicaray, Margaret J Wright, Meredith N Braskie, Gautam Prasad, Paul M Thompson
2017 IEEE 14th International Symposium on Biomedical Imaging (ISBI 2017); IEEE; pp. 961-965 (2017)

17. **Evaluating 35 methods to generate structural connectomes using pairwise classification**
Dmitry Petrov, Alexander Ivanov, **Joshua Faskowitz**, Boris Gutman, Daniel Moyer, Julio Villalon, Neda Jahanshad, Paul Thompson

International Conference on medical Image Computing and Computer-Assisted Intervention; Springer, Cham; pp. 515-522 (2017)

16. **Structural connectome validation using pairwise classification**
Dmitry Petrov, Boris Gutman, Alexander Ivanov, **Joshua Faskowitz**, Neda Jahanshad, Mikhail Belyaev, Paul Thompson
2017 IEEE 14th International Symposium on Biomedical Imaging (ISBI 2017); IEEE; pp. 451-455 (2017)
15. **Cortical connectome registration using spherical demons**
Dmitry Isaev, Boris A Gutman, Daniel Moyer, **Joshua Faskowitz**, Paul M Thompson
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14. **The core genetic network underlying sulcal morphometry**
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13. **Utilizing brain measures for large-scale classification of autism applying EPIC**
Marc B Harrison, Brandalyn C Riedel, Gautam Prasad, Neda Jahanshad, **Joshua Faskowitz**, Paul M Thompson
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12. **Topological Modules of Human Brain Networks Are Anatomically Embedded: Evidence from Modularity Analysis at Multiple Scales**
Anvar Kurmukov, Yulia Dodonova, Margarita Burova, Ayagoz Mussabayeva, Dmitry Petrov, **Joshua Faskowitz**, Leonid E Zhukov
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11. **Using Geometry of the Set of Symmetric Positive Semidefinite Matrices to Classify Structural Brain Networks**
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10. **Multi-modal registration improves group discrimination in pediatric traumatic brain injury**
Emily L Dennis, Faisal Rashid, Julio Villalon-Reina, Gautam Prasad, **Joshua Faskowitz**, Talin Babikian, Richard Mink, Christopher Babbitt, Jeffrey Johnson, Christopher C Giza, et al.
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9. **A continuous model of cortical connectivity**
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8. **Population learning of structural connectivity by white matter encoding and decoding**
Dajiang Zhu, Neda Jahanshad, Brandalyn C Riedel, Liang Zhan, **Joshua Faskowitz**, Gautam Prasad, Paul M Thompson
2016 IEEE 13th International Symposium on Biomedical Imaging (ISBI); IEEE; pp. 554-558 (2016)
7. **Genetic analysis of cortical sulci in 1,009 adults**
Fabrizio Pizzagalli, Guillaume Auzias, Peter Kochunov, **Joshua I Faskowitz**, Katie L McMahon, Greig I De Zubicaray, Nicholas G Martin, Margaret J Wright, Neda Jahanshad, Paul M Thompson
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6. **Comparison of template registration methods for multi-site meta-analysis of brain morphometry**
Joshua Faskowitz, Greig I de Zubicaray, Katie L McMahon, Margaret J Wright, Paul M Thompson, Neda Jahanshad
Medical Imaging 2016: Biomedical Applications in Molecular, Structural, and Functional Imaging; International Society for Optics and Photonics; Vol. 9788, pp. 978822 (2016)

5. **Embedded sparse representation of fMRI data via group-wise dictionary optimization**
   Dajiang Zhu, Binbin Lin, **Joshua Faskowitz**, Jieping Ye, Paul M Thompson
   Medical Imaging 2016: Image Processing; International Society for Optics and Photonics; Vol. 9784, pp. 97841K (2016)
4. **The heritability of the functional connectome is robust to common nonlinear registration methods**
   George W Hafzalla, Gautam Prasad, Vatche G Baboyan, **Joshua Faskowitz**, Neda Jahanshad, Katie L McMahon, Greig I de Zubicaray, Margaret J Wright, Meredith N Braskie, Paul M Thompson
   Medical Imaging 2016: Image Processing; International Society for Optics and Photonics; Vol. 9784, pp. 97841R (2016)
3. **Blockmodels for connectome analysis**
   Daniel Moyer, Boris Gutman, Gautam Prasad, **Joshua Faskowitz**, Greg Ver Steeg, Paul Thompson
   11th International Symposium on Medical Information Processing and Analysis; International Society for Optics and Photonics; Vol. 9681, pp. 96810A (2015)
2. **Genetic analysis of structural brain connectivity using DICCCOL models of diffusion MRI in 522 twins**
   Dajiang Zhu, Liang Zhan, **Joshua Faskowitz**, Madelaine Daianu, Neda Jahanshad, Greig I De Zubicaray, Katie L McMahon, Nicholas G Martin, Margaret J Wright, Paul M Thompson
   2015 IEEE 12th International Symposium on Biomedical Imaging (ISBI); IEEE; pp. 1167-1171 (2015)
1. **Heritability of brain network topology in 853 twins and siblings**
   Liang Zhan, Neda Jahanshad, **Joshua Faskowitz**, Dajiang Zhu, Gautam Prasad, Nicholas G Martin, Greig I de Zubicaray, Katie L McMahon, Margaret J Wright, Paul M Thompson
   2015 IEEE 12th International Symposium on Biomedical Imaging (ISBI); IEEE; pp. 449-453 (2015)

Conference Posters (first-author only)

21. **Spatial Similarity Maps Across the Cortex**
    Joshua Faskowitz, Peter A. Bandettini, Javier Gonzalez-Castillo
    Poster and talk presented at OHBM, Bordeaux, Fr (2026-06)
20. **Edge event organization across temporal categories**
    Joshua Faskowitz, Javier Gonzalez-Castillo, Peter A. Bandettini
    Poster accepted for OHBM, Brisbane, AUS (2025-05)
19. **On the features of spiking connectivity**
    Joshua Faskowitz, Javier Gonzalez-Castillo, Daniel A. Handwerker, Peter A. Bandettini
    Poster presented at OHBM, Seoul, KOR (2024-06)
18. **On the static and dynamic features of edge time series**
    Joshua Faskowitz, Tyler Morgan, Daniel A. Handwerker, Javier Gonzalez-Castillo, Peter A. Bandettini
    Poster presented at OHBM, Montreal, CA (2023-07)
17. **Mammalian brain network distances and their relationship to taxonomy and phylogeny**
    Joshua Faskowitz, Maria Grazia Puxeddu, Martijn van den Heuvel, Bratislav Misic, Yossi Yovel, Yaniv Assaf, Richard Betzel, Olaf Sporns
    Poster presented at SfN, San Diego, U.S.A. (2022-11)
16. **An edge-centric model for harmonizing multi-relational network datasets**
    Joshua Faskowitz, Jacob Tanner, Bratislav Misic, Richard Betzel
    Poster presented virtually at OHBM (2021-06)
15. **Edge community structure of functional MRI and meta-analytic activation**
    Joshua Faskowitz, Thomas Varley, Richard Betzel, Olaf Sporns
    Poster presented virtually at OHBM (2021-06)
14. **The brainlife.io cloud-services for functional network neuroscience**
    Joshua Faskowitz, Conner Victroy, David Hunt, Franco Delogu, Soichi Hayashi, Richard Betzel, Franco Pestilli
    Poster presented virtually at OHBM, Montreal, CA (2020-06)

13. **Edge functional connectivity reveals overlapping community structure**
Joshua Faskowitz, Farnaz Zamani Esfahlani, Youngheun Jo, Olaf Sporns, Richard F Betzel
Poster presented virtually at OHBM, Montreal, CA (2020-06)

12. **The edge-centric representation of functional brain networks**
Joshua Faskowitz, Youngheun Jo, Farnaz Zamani Esfahlani, Olaf Sporns, Richard F Betzel
Poster presented virtually at OHBM, Montreal, CA (2020-06)

11. **Mapping the Community Structure of the Connectome with Weighted Stochastic Block Modeling**
Joshua Faskowitz, Olaf Sporns
Poster presented at OHBM, Rome, IT (2019-06)

10. **Analyzing the Structure of Brain Networks using Stochastic Block Models**
J Faskowitz, O Sporns
Poster presented at NetSci and Network Neuroscience satellite, Burlington, U.S.A. (2019)

9. **Weighted stochastic blockmodels of the human connectome across the life span**
Joshua Faskowitz, Xiaoran Yan, Xi-Nian Zuo, Olaf Sporns
Poster presented at SfN, San Diego, U.S.A (2018)

8. **Development of Community Structure in the Human Connectome across the Life Span: An Application of Weighted Stochastic Blockmodels**
Joshua Faskowitz, Xiaoran Yan, Xi-Nian Zuo, Olaf Sporns
Poster presented at NetSci and Network Neuroscience satellite, Indianapolis, U.S.A (2017)

7. **Cortical investigation of bipolar disorder reveals inferior frontal gyral and sulcal abnormalities**
Joshua Faskowitz, Katie McMahon, Greig de Zubicaray, P. M. Thompson, Margaret Wright, Neda Jahanshad
Poster presented at OHBM, Geneva, CH (2016)

6. **Cortical investigation of bipolar disorder reveals inferior frontal gyral and sulcal abnormalities**
J. Faskowitz, F. Pizzagalli, Neda Jahanshad, Christopher Ching, B. Mwangi, J. C. Soares, P. M. Thompson
Poster presented at OHBM, Geneva, CH (2016)

5. **Cortical abnormalities in patients with bipolar disorder more localized than in those with schizophrenia**
J. Faskowitz, F. Pizzagalli, B. Mwangi, P. Kochunov, P. M. Thompson, J. C. Soares, Jahanshad N.
Poster presented at SfN, San Diego, U.S.A (2016)

4. **Test-retest reliability of cortical parcellations in 165 healthy adults for multi-site analyses in the ENIGMA consortium**
J Faskowitz, H Hibar, P. M. Thompson, N Jahanshad
Poster presented at SfN, Chicago, U.S.A (2015)

3. **CD56+ Natural Killer cell counts associate with reductions in white matter fractional anisotropy**
Joshua Faskowitz, Neda Jahanshad, NK Hansell, Greig de Zubicaray, Katie McMahon, N Martin, Margaret Wright, P. M. Thompson
Poster presented at OHBM, Honolulu, U.S.A (2015)

2. **Brain white matter integrity in bipolar disorder subtypes assessed with diffusion tensor imaging**
J. Faskowitz, Christopher Ching, J. C. Soares, P. M. Thompson, Neda Jahanshad
Poster presented at Cognitive Neuroscience Society, San Francisco, U.S.A (2015)

1. **The influence of emotion on the recognition of change**
J Faskowitz, A Ponzio, J. J. Castrellon, M Mather
Poster presented at Western Psychological Association, Portland, U.S.A (2014)